14

JEE RANKERS TEST SERIES-2026

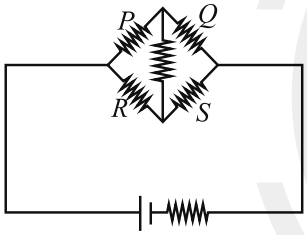
Important Instructions:

1. The test is of **3 hours** duration.
2. The test booklet consists of **75** questions. The maximum marks are **300**.
3. There are three Sections in the question paper, Section I, II & III consisting of Section-I **(Physics)**, Section-II **(Chemistry)**, Section-III **(Mathematics)** and having **25 questions** in each part in which first **20 questions** are **Multiple Choice Questions (MCQs)** with **only one correct** option and last **5 questions** are **Numerical Value Type**.
4. Each question carries 4 marks. For each correct response, the candidate will get **4 marks**. For each incorrect response, **one** mark will be deducted from the total scores. No marks will be awarded for unattempted questions.
5. No student is allowed to carry any textual material, printed or written, bits of papers, paper, mobile phone, any electronic device, etc. inside the examination room/hall.

PHYSICS

Single Correct Type Questions

1. The resistance of P, Q, R, S arms of a Wheatstone bridge are 5, 15, 20 and 60 Ω. A cell of 4 volt emf and 4 internal resistance is connected with them, then the current flowing (in ampere) from battery is



(1) 0.1 A (2) 0.2 A

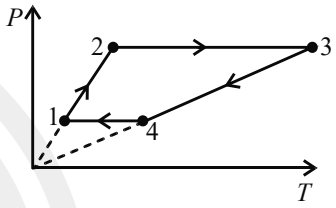
(3) 1 A (4) 2 A

2. A sphere of solid material of specific gravity 8 has a concentric spherical cavity and just sinks in water. The ratio of radius of cavity to that of outer radius of the sphere must be

(1) $\frac{7^{1/3}}{2}$ (2) $\frac{5^{1/3}}{2}$

(3) $\frac{9^{1/3}}{2}$ (4) $\frac{3^{1/3}}{2}$

3. Three moles of an ideal monoatomic gas perform a cycle shown in figure. The gas temperatures in different states are T_1 = 400 K, T_2 = 800 K, T_3 = 2400 K, and T_4 = 1200 K. The work done by the gas during the cycle is: $\left[R = \frac{25}{3}\text{J/mol} - \text{k}\right]$



(1) 5 kJ (2) 10 kJ

(3) 15 kJ (4) 20 kJ

4. The electric charge in uniform motion produces-

(1) an electric field only

(2) a magnetic field only

(3) both electric and magnetic fields

(4) neither electric nor magnetic fields

5. **Statement-I:** When a Si sample is doped with Boron, it becomes P type and when doped by Arsenic it becomes N-type semi-conductor such that P-type has excess of holes and N-type has excess of electrons.

Statement-II: When such P-type ands N-type semiconductors, are fused to make a junction, a current will automatically flow which can be detected with an externally connected ammeter.

In light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement-I and statement-II are incorrect

(2) Statement-I is incorrect but statement-II is correct

(3) Both Statement-I and statement-II are correct

(4) Statement-I is correct but statement-II is incorrect

6. **Statement-1:** In spontaneous fission, the energy is always released.

Statement-2: Spontaneous fission occurs to lower the binding energy of reactant nuclei.

(1) Statement-1 is True, Statement-2 is True; Statement-2 **is** a correct explanation for Statement-1

(2) Statement-1 is True, Statement-2 is True; Statement-2 **is NOT** a correct explanation for Statement-1

(3) Statement-1 is True, Statement-2 is False

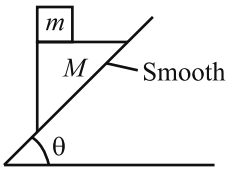
(4) Statement-1 is False, Statement-2 is True.

7. Two bar magnets of the same length and breadth but having magnetic moment M and 2M are joined together pole for pole and suspended by a string. The time of oscillation of this assembly in a magnetic field of strength B is 3 sec. What will be the period of oscillation if the polarity of one of the magnets is changed and the combination is again made to oscillate in the same field?

(1) $3\sqrt{3}$ (2) 3

(3) $\sqrt{3}$ (4) 6

8. A triangular prism of mass M with a block of mass m placed on it is released from rest on a smooth inclined plane of inclination θ. The block does not slip on the prism. Then



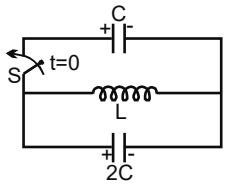
(1) The acceleration of the prism is $g \cos \theta$.

(2) The acceleration of the prism is $g \tan \theta$.

(3) The minimum coefficient of friction between the block and the prism is $\mu_{min} = \cot \theta$.

(4) The minimum coefficient of friction between the block and the prism is $\mu_{min} = \tan \theta$.

9. In the given LC circuit if initially capacitor C has charge Q on it and $2C$ has charge $2Q$ .The polarities are as shown in the figure. Then after closing switch S at $t = 0$



(1) energy will get equally distributed in both the capacitor just after closing the switch.

(2) initial rate of growth of current in inductor will be $2Q/3CL$

(3) maximum energy in the inductor will be $3Q^2/2C$

(4) None of these

10. Given below are two statements: one is labelled as Assertion (A) and other is labelled as Reason (R).

Assertion (A): The photoelectric effect does not takes place, if the energy of the incident radiation is less than the work function of a metal.

Reason (R): Kinetic energy of the photoelectrons is zero, if the energy of the incident radiation is equal to the work function of a metal.

In the light of the above statements, choose the most appropriate answer from the options given below.

(1) Both A and R are correct and R is the correct explanation of A

(2) Both A and R are correct but R is not the correct explanation of A

(3) A is correct but R is not correct

(4) A is not correct but R is correct

11. A block of mass m is attached to one end of a massless spring of spring constant k. The other end of spring is fixed to a wall. The block can move on a horizontal rough surface. The coefficient of friction between the block and the surface is μ. Then the compression of the spring for which maximum extension of the spring becomes half of maximum compression is

(1) $\frac{2mg\mu}{k}$ (2) $\frac{mg\mu}{k}$

(3) $\frac{4mg\mu}{k}$ (4) None of these

12. A uniform rod AB of mass m and length l is at rest on a smooth horizontal surface. A impulse J is applied to the end B perpendicular to the rod in horizontal direction. Speed of particle P at a

distance $\frac{l}{6}$ from the centre towards A of the rod after time $t = \frac{\pi m l}{12J}$ is:

(1) $2\frac{J}{m}$ (2) $\frac{J}{\sqrt{2}m}$

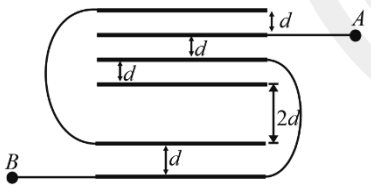
(3) $\frac{J}{m}$ (4) $\sqrt{2}\frac{J}{m}$

13. The instantaneous velocity of a particle moving in a straight line is given as $v = \alpha t + \beta t^2$, where α and β are constants. The distance travelled by the particle between 1s and 2s is:

(1) $\frac{\alpha}{2}+\frac{\beta}{3}$ (2) $\frac{3}{2}\alpha+\frac{7}{3}\beta$

(3) $\frac{3}{2}\alpha+\frac{7}{2}\beta$ (4) $3\alpha+7\beta$

14. Find equivalent capacitance between points A and B. Consider each conducting plate having same dimensions, each of area A with separation between adjoining plates as mentioned in figure and neglect the thickness of the plate. Consider $C = \frac{\varepsilon_0 A}{d} = 7\mu F$, where A is area of plates



(1) 7 μF (2) 11 μF

(3) 12 μF (4) 15 μF

15. If the period of revolution of an artificial satellite just above the earth's surface is T and the density of earth is ρ, then ρT^2. (G = universal gravitational constant)

(1) Is a universal constant whose value is $\frac{3\pi}{G}$

(2) Is a universal constant whose value is $\frac{3\pi}{2G}$

(3) Is proportional to radius of earth R

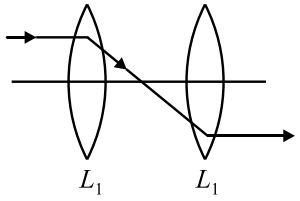
(4) Is proportional to square of the radius of earth R^2 Here G = universal gravitational constant

16. Visible light passing through a circular hole form a diffraction disc of radius 0.1 mm on a screen. If X-ray is passed through the same set-up, the radius of the diffraction disc will be:

(1) Zero (2) < 0.1 mm

(3) 0.1 mm (4) > 0.1 mm

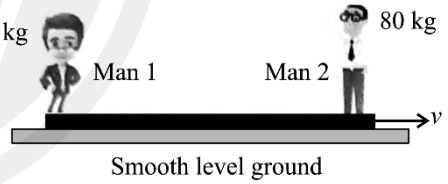
17. In the figure given below, there are two convex lens L_1 and L_2 having focal length of f_1 and f_2 respectively. The distance between L_1 and L_2 will be



(1) f_1 (2) f_2

(3) $f_1 + f_2$ (4) $f_1 - f_2$

18. Two men of mass 60 kg and 80 kg stand on a plank of mass 20 kg. Both of them can jump with a velocity of 1 m/s relative to the plank. In each event shown in column I, find the speed of plank after the event



Column-I		Column-II	
I	Man '1' alone jumps to the left	P	$\frac{17}{40}$ m/s
II	Man '2' alone jumps to the right	Q	$\frac{1}{2}$ m/s
III	Man '1' jumps to left and Man '2' jumps to right simultaneously	R	$\frac{3}{8}$ m/s
IV	Man '1' jumps to left and after that Man '2' jumps to right	S	$\frac{1}{8}$ m/s

(1) I→Q; II→P; III→R; IV→S

(2) I→R; II→Q; III→S; IV→P

(3) I→Q; II→R; III→S; IV→Q

(4) I→R; II→R; III→Q; IV→P

19. **Assertion:** The wave function of a pulse given by $y = \frac{3}{(2x+3t+4)}$ propagates in (– ve x-direction)

Reason: The given wave function is of the form $y = f(kx+\omega t)$ which represent a wave travelling in negative x- direction. (symbols have usual meaning)

(1) Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are true, but Reason is not correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Assertion is false but Reason is true.

20. A particle is projected with velocity u horizontally from the top of a smooth sphere of radius a so that it slides down the outside of the sphere. If the particle leaves the sphere when it has fallen a height $a/4$, the value of u is (g = acceleration due to gravity)

(1) $\sqrt{ag}$ (2) $\frac{\sqrt{ag}}{4}$

(3) $\frac{\sqrt{ag}}{2}$ (4) $\frac{\sqrt{ag}}{3}$

Integer Type Questions

21. A transverse wave of amplitude 0.50 mm and frequency 100Hz is produced on a wire stretched to a tension of 100 N If the wave speed is 100 m/s, what average power (in mW) is the source transmitting to the wire?

(Take $\pi^2 = 10$)

22. A 100 kg mass, fastened to the end of a steel wire of unstretched length 1.0m, is whirled in a vertical circle with a speed of 5 m/s at the bottom of the circle. The cross-sectional area of wire is 0.070 cm^2. The elongation of the wire when the mass is at the lowest point of its path. ($Y_{steel} = 2\times10^{10}\frac{N}{m^2}$) is 25×10^{-x} m. Find the value of x (g = 10 m/s^2)

23. Initially a gas of diatomic molecules is contained in a cylinder of volume V_1 at a pressure P_1 and temperature 250 K. Assuming that 25% of the molecules get dissociated causing a change in number of moles. The pressure of the resulting gas at temperature 2000 K, when contained in a volume $2V_1$ is given by P_2. The ratio $\frac{P_2}{P_1}$ is:

24. A satellite of mass m initially at rest on earth surface is launched into a circular orbit of double the radius of earth. The radius of earth is R_e. The minimum energy required to do so is $\frac{x}{4}mgR_e$ find x. (g = acceleration due to gravity)

25. A pulley of radius 1.5 m is rotated about its axis by a force $F = (12t - 3t^2)$ N applied tangentially (while t is measured in seconds). If moment of inertia of the pulley about its axis of rotation is 4.5 kg m^2, the number of rotations made by the pulley before its direction of motion is reversed, will be $\frac{K}{\pi}$. The value of K is

CHEMISTRY

Single Correct Type Questions

26. **Assertion (A):** Entropy change in reversible adiabatic expansion of an ideal gas is zero.

Reason (R): The increase in entropy due to volume increase just compensate the decrease in entropy due to fall in temperature.

(1) Both A and R are correct; and the R is the correct explanation for the A

(2) Both A and R are correct; but the R is not the correct explanation for the A.

(3) A is incorrect, R is correct.

(4) Both the A and R are incorrect.

27. For an octahedral complex, which of the following d electron configuration will give maximum crystal-field stabilisation energy

(1) High spin d^6 (2) Low spin d^4

(3) Low spin d^5 (4) High spin d^7

28. A 0.095 gm sample of purified carbonate was dissolved in 50 ml of 0.1 M HCl and boiled to eliminate CO_2. Back titration of excess HCl require 24.2 ml of 0.1 M NaOH. Identify carbonate.

(1) Na_2CO_3 (2) Li_2CO_3

(3) $CaCO_3$ (4) $MgCO_3$

29. Clemmensen reduction of a ketone is carried out in the presence of which of the following?

(1) Glycol with KOH

(2) Zn-Hg with HCl

(3) $LiAlH_4$

(4) H_2 and Pt as catalyst

30. When a lead storage battery is discharged

(1) $PbSO_4$ is formed

(2) Pb is formed

(3) SO_2 is consumed

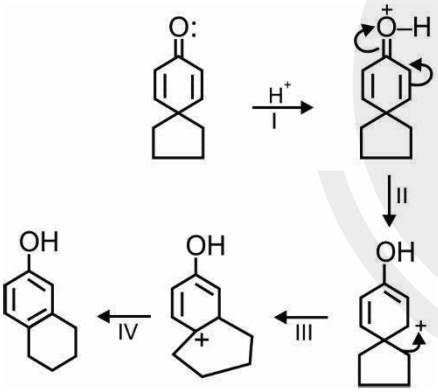
(4) H_2SO_4 is formed

31. Among the following metals, which has highest melting point?

(1) Ti (2) Cr

(3) Fe (4) Cu

32. Consider the following mechanism.



Identify the correct statement.

(1) Step-II is the RDS of the reaction

(2) Step-III is the RDS of the reaction

(3) Overall ΔH of the reaction is positive

(4) In step-III, the degree of unsaturation is increased by one

33. Aluminium phosphate is 100% ionised in 0.01 molal aqueous solution. Hence, $\Delta T_b / K_b$ is:

(1) 0.01

(2) 0.015

(3) 0.0175

(4) 0.02

34. $PhNH_2$ and PhNHMe can be differentiated by:

(1) $CHCl_3$,KOH

(2) $NaNO_2$,HCl

(3) Hinsberg reagent

(4) All of these

35. The wavelength (λ) of monochromatic light coming from some light sources is listed below. How many of these sources will be able to exhibit photoelectric effect if incident upon surface of Li metal (work function, $\phi = 2.4$ eV).

Light Source	λ (nm)
A	100
B	200
C	300
D	400
E	500
F	600
G	700
H	800
I	900

(1) 6 (2) 7

(3) 8 (4) 5

36. Consider the following four elements, which are represented according to long form of periodic table.

	Y	
W	X	Z

Here W, Y and Z are left, up and right elements with respect to the element 'X' and 'X' belongs to 16th group and 3rd period. Then according to given information the incorrect statement regarding given elements is :

(1) Maximum electronegativity : Y

(2) Maximum catenation tendency : X

(3) Maximum Electron affinity : Z

(4) Y exhibits variable covalency

37. Which of the following amino acid is basic amino acid?

(1) Leucine (2) Proline

(3) Aspargine (4) Lysine

38. n_{factor} of $Fe_{0.9}O$ in following reaction is :

$$H^+ + Fe_{0.9}O + MnO_4^- \rightarrow Mn^{2+} + Fe_2O_3 + H_2O$$

(1) $\frac{7}{10}$ (2) $\frac{20}{9}$

(3) $\frac{27}{9}$ (4) $\frac{7}{9}$

39. Match Column-1 with Column-2 and select the correct answer using the code given below the columns:

Column-1		Column-2	
I	If the activation energy is 65 kJ then how much time faster a reaction proceed at 25°C than at 0°C (if ln 11 = 2.4)	P	2
II	Rate constant of a first - order reaction is 0.0693 min^{-1}. If we start with 20 mol L^{-1}, it is reduced to 2.5 mol L^{-1} in how many minutes	Q	Zero
III	Half - lives of first - order and zero order reactions are same. Ratio of rate constants for first order by zero order at the start of reaction is how many times of 0.693. Assume initial concentration to be unit for the both.	R	11
IV	The half-life periods are given, $[A]_0$ 0.0677 0.136 $t_{1/2}$ 240 480 Order of the reaction is:	S	30

(1) I→R; II→S; III→ P; IV→Q

(2) I→S; II→R; III→P; IV→Q

(3) I→R; II→P; III→S; IV→Q

(4) I→R; II→S; III→Q; IV→P

40. The oxide of a metal (R) can be reduced by the metal (P) and metal (R) can reduce the oxide of metal (Q). Then the decreasing order of the reactivity of metal (P), (Q) and (R) with oxygen is

(1) P > Q > R (2) P > R > Q

(3) R > P > Q (4) Q > P > R

41. pH of 3×10^{-3} M solution of H_3X will be (assume $\alpha_1 = 1$, $\alpha_2 = 1/3$, α_3 = negligible)

(1) 2.40 (2) 3.0

(3) 3.4771 (4) 4.0

42. $CH_3-CH_2-\underset{\underset{O}{\|}}{C}-CH_3 \xrightarrow[(ii)\ CH_3MgBr]{(i)\ CH_3MgBr}$

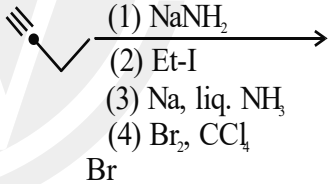
Product is:

(1) $CH_3-CH_2-\overset{\overset{CH_3}{|}}{\underset{\underset{OH}{|}}{C}}-CH_3$

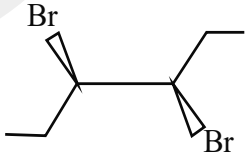
(2) $CH_3-CH_2-\overset{\overset{CH_3}{|}}{CH}-\underset{\underset{OH}{|}}{CH_2}$

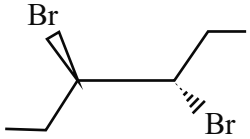
(3) $CH_3-CH_2-\underset{\underset{OH}{|}}{CH}-CH_3$

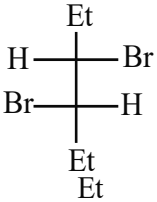
(4) $CH_3-CH_2-\underset{\underset{CH_3}{|}}{CH}-CH_3$

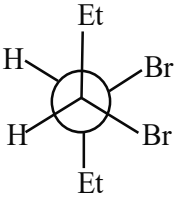
43. 

major product is?

(1) 

(2) 

(3) 

(4) 

44. The magnetic moment of a complex ion is 2.83 BM. The complex ion is

(1) $[V(H_2O)_6]^{3+}$ (2) $[Cr(H_2O)_6]^{3+}$

(3) $[Cu(CN)_4]^{2-}$ (4) $[MnCl_4]^{2-}$

45. The dipole moment of CH_3X (where X is a halogen) following the order

(1) $CH_3F > CH_3Cl > CH_3Br$

(2) $CH_3F > CH_3Cl < CH_3Br$

(3) $CH_3F < CH_3Cl > CH_3Br$

(4) $CH_3F < CH_3Cl < CH_3Br$

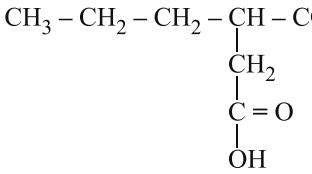
Integer Type Questions

46. 0.117 g of an organic compound on heating with conc. HNO_3 and silver nitrate in carius furnace gave 0.42 g of AgCl. Find the percentage of chlorine in the compound. (Nearest integer)

47. The rate constant for the forward reaction A (g) $\rightleftharpoons$ 2B(g) is 1.5×10^{-3} s^{-1} at 100 K. If 10^{-5} moles of A and 100 moles of B are present in a 10 litre vessel at equilibrium then rate constant for the backward reaction at this temperature is $x \times 10^{-12}$ The value of x is___

48. A current of 0.1 amp is passed though excess of molten mixture of Al_2O_3 and Na_3AlF_6 for 96.5 hours. The mass of Al (in gm) deposited at the cathode, with $\frac{1000}{10.8}\%$ current efficiency is (Al = 27).

49.



Number of carbon in selected principal carbon chain for IUPAC nomenclature will be

50. A vessel of 250 litre was filled with 0.01 mole of Sb_2S_3 and 0.01 mole of H_2 to attain the equilibrium at 440°C as

Sb_2S_3 (s) + $3H_2$ (g) $\rightleftharpoons$ 2Sb (s) + $3H_2S$ (g)

After equilibrium, the H_2S formed was analysed by dissolving it in water and treating with excess of Pb^{2+} to give 1.19 g of PbS as precipitate. What is value of K_c of the reaction at 440°C? (At. Mass: Pb = 207, S = 32)

MATHEMATICS

Single Correct Type Questions

51. Let the maximum area of the triangle that can be inscribed in the ellipse $\frac{x^2}{a^2}+\frac{y^2}{4}=1, a>2$, having one of its vertices at one end of the major axis of the ellipse and one of its sides parallel to the y-axis, be $6\sqrt{3}$. Then the eccentricity of the ellipse is:

(1) $\frac{\sqrt{3}}{2}$ (2) $\frac{1}{2}$

(3) $\frac{1}{\sqrt{2}}$ (4) $\frac{\sqrt{3}}{4}$

52. Suppose that a function $f: R \to R$ satisfies $f(x + y) = f(x) \cdot f(y)$ for all $x, y \in R$ and $f(1) = 3$. If $\sum_{i=1}^{n} f(i) = 363$, then $2n$ is equal to

(1) 10

(2) 11

(3) 9

(4) 13

53. If p & q are distinct reals, then $2\{(x-p)(x-q) + (p-x)(p-q) + (q-x)(q-p)\} = (p-q)^2 + (x-p)^2 + (x-q)^2$ is satisfied by:

(1) no values of 'x'

(2) exactly one value of 'x'

(3) exactly two values of 'x'

(4) infinite values of 'x'

54. Let a and b be two consecutive integers selected from the first 20 natural numbers. The probability that $\sqrt{a^2+b^2+a^2b^2}$ is an odd positive integer is

(1) $\frac{9}{19}$

(2) $\frac{10}{19}$

(3) $\frac{13}{19}$

(4) 1

55. $100\tan\left(3\tan^{-1}\left(\frac{1}{2}\right)+2\cos^{-1}\left(\frac{1}{\sqrt{5}}\right)\right)+$ $4\sqrt{2}\tan\left(\frac{1}{2}\tan^{-1}(2\sqrt{2})\right)$ is equal to.

(1) 29 (2) 54
(3) 60 (4) 38

56. Match the items of Column-1 with those of Column-2.

Column-1		Column-2	
I	$\int\frac{\cos 5x+\cos 4x}{1-2\cos 3x}dx=$	P	$\frac{1}{6}\left(\sec^3 2x-3\sec 2x\right)$
II	$\int\frac{\cos 7x-\cos 8x}{1+2\cos 5x}dx=$	Q	$\frac{2}{9}(2+\sin 3x)^{3/2}$
III	$\int\tan^3 2x\sec 2x dx=$	R	$\frac{-1}{2}(2\sin x+\sin 2x)$
IV	$\int\sqrt{2+\sin 3x}\cos 3x dx=$	S	$\frac{1}{2}\sin 2x$ $-\frac{1}{3}\sin 3x+c$

(1) I→R; II→S; III→P: IV→Q
(2) I→P; II→S; III→P: IV→Q
(3) I→Q; II→S; III→Q: IV→P
(4) I→Q; II→P; III→P: IV→P

57. A vector $\vec{a}$ has components $2p$ and 1 with respect to a rectangular cartesian system. The system is rotated through a certain angle about the origin in the counterclockwise sense. If with respect to the new system $\vec{a}$, has components $p + 1$ and 1, then

(1) $p = 0$
(2) $p = 1$ or $p = -\frac{1}{3}$
(3) $p = -1$ or $p = \frac{1}{3}$
(4) $p = 1$ or $p = -1$

58. Let $S=\left\{z\in C:\bar{z}=i\left(z^2+\text{Re}\left(\bar{z}\right)\right)\right\}$. Then $\prod_{z\in S}|z|^2$ is equal to (Re(z) = real part of z)

(1) $\frac{7}{2}$ (2) 0
(3) $\frac{5}{2}$ (4) 3

59. If $AB = 0$ for the matrices

$A=\begin{bmatrix}\cos^2\theta & \cos\theta\sin\theta\\ \cos\theta\sin\theta & \sin^2\theta\end{bmatrix}$ and

$B=\begin{bmatrix}\cos^2\phi & \cos\phi\sin\phi\\ \cos\phi\sin\phi & \sin^2\phi\end{bmatrix}$ then $\theta - \phi$ is

(1) an odd multiple of $\frac{\pi}{2}$
(2) an odd multiple of π
(3) an even multiple of $\frac{\pi}{2}$
(4) 0

60. A person writes letters to five friends and addresses on the corresponding envelopes. In how many ways can the letters be placed in the envelopes so that at least four of them are in the wrong envelopes?

(1) 89 (2) 40
(3) 44 (4) 109

61. For each positive integer n, define $f_n(x)=\text{Min}\left(\frac{x^n}{n!},\frac{(1-x)^n}{n!}\right)$ for $0 \le x \le 1$. Let $I_n=\int_0^1 f_n(x)dx$ for n ≥ 1. Then the value of $\sum_{n=1}^{\infty} I_n$ is

(1) $2\sqrt{e}-3$
(2) $2\sqrt{e}-2$
(3) $2\sqrt{e}-1$
(4) $2\sqrt{e}$

62. If S be such that $S = {}^nC_1 - 3{}^nC_3 + 3^2.{}^nC_5 - 3^3.{}^nC_7 \ldots$; then S is equal to

(1) $(-1)^n\cdot\frac{2^n}{\sqrt{3}}\sin\frac{2n\pi}{3}$
(2) $(-1)^{n+1}\cdot\frac{2^n}{\sqrt{3}}\sin\frac{2n\pi}{3}$
(3) $(-1)^{n+1}\cdot\frac{2^n}{\sqrt{3}}\cos\frac{2n\pi}{3}$
(4) $(-1)^n\cdot\frac{(2)^n}{\sqrt{3}}\cos\frac{2n\pi}{3}$

63. If the lines $\frac{x}{1}=\frac{y}{2}=\frac{z}{3}$, $\frac{x-1}{3}=\frac{y-2}{-1}=\frac{z-3}{4}$ and $\frac{x+k}{3}=\frac{y-1}{2}=\frac{z-2}{h}$ are concurrent then

(1) $h=-2, k=-6$

(2) $h=\frac{1}{2}, k=2$

(3) $h=6, k=2$

(4) $h=2, k=\frac{1}{2}$

64. Let

$$x=\sqrt{1+\frac{1}{1^2}+\frac{1}{2^2}}+\sqrt{1+\frac{1}{2^2}+\frac{1}{3^2}}+\sqrt{1+\frac{1}{3^2}+\frac{1}{4^2}}$$

$$+\cdots+\sqrt{1+\frac{1}{2009^2}+\frac{1}{2010^2}}$$

Then $\frac{(2010)x-2009}{2010}=$

(1) 2010 (2) 2009

(3) 1999 (4) 2000

65. A rectangle $ABCD$ is inscribed in the circle $x^2 + y^2 + 3x + 12y + 2 = 0$. If the co-ordinates of A and B are (3, –2) and (–2, 0), then the other two vertices of the rectangle are (a, b) and (c, d) then the value of $|a + b + c + d|$ is:

(1) 29 (2) 15

(3) 23 (4) 55

66. The ends of the rod of length ℓ moves on two mutually perpendicular lines, find the locus of the point on the rod which divides it in the ratio $m_1 : m_2$

(1) $m_1^2x^2+m_2^2y^2=\frac{\ell^2}{(m_1+m_2)^2}$

(2) $(m_2x)^2+(m_1y)^2=\left(\frac{m_1m_2\ell}{m_1+m_2}\right)^2$

(3) $(m_1x)^2+(m_2y)^2=\left(\frac{m_1m_2\ell}{m_1+m_2}\right)^2$

(4) none of these

67. Let [ ] represents the greatest integer function $\left[\cos^{-1}\sin^{-1}\tan^{-1}x\right]=1$ then 'x' lies in the interval.

(1) $[\sin\tan\cos 2, \tan\sin\cos 1]$

(2) $[\tan\sin\cos 2, \tan\sin\cos 2]$

(3) $[\tan\sin\cos 1, \tan\sin\cos 2]$

(4) $[\tan\sin\cos 2, \tan\sin\cos 1]$

68. The solution of the differential equation $\frac{\sqrt{x}dx+\sqrt{y}dy}{\sqrt{x}dx-\sqrt{y}dy}=\sqrt{\frac{y^3}{x^3}}$, is given by (where c is arbitrary constant)

(1) $\frac{3}{2}\log\left(\frac{y}{x}\right)+\log\left|\frac{x^{3/2}+y^{3/2}}{x^{3/2}}\right|$

$+\tan^{-1}\left(\frac{y}{x}\right)^{3/2}+c=0$

(2) $\frac{3}{2}\log\left(\frac{y}{x}\right)+\log\left|\frac{x^{3/2}+y^{3/2}}{x^{3/2}}\right|+\tan^{-1}\frac{y}{x}+c=0$

(3) $\frac{2}{3}\left(\frac{y}{x}\right)+\log\left|\frac{x+y}{x}\right|+\tan^{-1}\left(\frac{y^{3/2}}{x^{3/2}}\right)+c=0$

(4) $\frac{1}{2}\log\left(x^3+y^3\right)+\tan^{-1}\left(\frac{y}{x}\right)^{3/2}+c=0$

69. If A and B are two matrices such that $AB = B$ and $BA = A$, then

(1) $(A^5 + B^5)^3 = 2A - 2B$

(2) $(A^5 - B^5) = 2A^3 + 2B^3$

(3) $A - B$ is idempotent

(4) $A - B$ is nilpotent

70. If $xy = ax^2 + (b/x)$, then $x\frac{d^2y}{dx^2}+2\frac{dy}{dx}=$

(1) $\frac{y}{x}$

(2) $\frac{-y}{x}$

(3) $\frac{2y}{x}$

(4) $\frac{-2y}{x}$

Integer Type Questions

71. Let $f(x)$ is a function satisfying the condition $f(x)=\frac{1}{3}\left[f(x+6)+\frac{6}{f(x+7)}\right]$ and $f(x) > 0 \ \forall \ x \in R$ then if $\lim_{x\to\infty} f(x)=\sqrt{m}$ then the value of m is _____.

72. Consider the 8 × 8 square matrix filled with the natural numbers from 1 to 64 as is given below.

$$\begin{bmatrix} 1 & 2 & 3 & 4 & \cdots & 8 \\ 9 & 10 & 11 & 12 & \cdots & 16 \\ 17 & 18 & 19 & 13 & \cdots & 24 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 57 & 58 & 59 & 60 & \cdots & 64 \end{bmatrix}$$

A number is selected from the board and the corresponding row and column are deleted. Again another number is selected and the row and column are deleted. The process is continued upto 8 times so that no row and no column is left. Then the sum of the numbers so selected is_____.

73. The area of the closed figure bounded by $y = x$, $y = -x$ & the tangent to the curve $y = \sqrt{x^2-5}$ at the point (3, 2) is:

74. If $\sum_{i=1}^{11}(x_i-4)=11$ and $\sum_{i=1}^{11}(x_i-4)^2=44$ then the variance of $x_1, x_2, x_3 --, x_{11}$ is

75. Length of the chord intercepted by the parabola $y = x^2 + 3x$ on the line $x + y = 5$ is $m\sqrt{2}$, then find m.